GoPro + Garmin Edge Bike Remote — Hardware Notes

Companion document to `gopro_garmin_remote_specs.md`, focused on the hardware/electronics side (PCB, power, programming, enclosure-adjacent electrical decisions). Same conventions as the specs document: sourced claims are marked, unverified/assumed points are flagged as such.

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1. Hardware summary

- **MCU** / **radio**: Nordic nRF52 with BLE + ANT+ on one chip — see the specs document for why this family was chosen over alternatives (e.g. ESP32). **Module under revision (§9)**: the Ebyte E73-2G4M08S1E previously listed here turned out to be an **nRF52833** (not supported by sdk-ant) and is dropped. Preferred replacement: **Fanstel BT832 (nRF52832, 14 × 16mm, castellated, in the JLCPCB library) + 32.768kHz crystal on our board**; alternatives Raytac MDBT42Q, E73-2G4M08S1C, E73-2G4M04S1B. Until §9 is closed, the nRF52832 figures quoted in this document are provisional.
- **Power**: single **CR2032** coin cell, directly powering the nRF52 (no regulator — native 1.7–3.6V operating range). A buffer capacitor (tantalum/polymer, 47–100µF, low ESR) absorbs BLE/ANT+ current spikes; a 100nF ceramic capacitor handles HF filtering. Battery level is read purely in software via the internal SAADC VDD channel — no extra BOM component.
- **Inputs**: 5 tact switches (individual GPIOs, no matrix), internal pull-up, active-low. Wake-up is GPIO PORT-event based (SENSE + LATCH), so idle current is dominated by leakage, not polling.
- **Sensing**: ambient temperature is read from the nRF52832's *internal* die-temperature peripheral — no external thermistor, no extra ADC input.
- **PCB**: custom board built around the E73 module's castellated footprint, with **extended pads** so it can be hand-soldered with an iron (no reflow oven required).
- **User feedback**: red + yellow-green (AlGaInP) indicator LEDs, resistor-fed straight from the cell and sunk by high-drive GPIOs (≈ 5mA red / ≈ 10mA green peak, short flashes) — see §8.
- **Programming/debug**: **SWD** (SWDIO / SWCLK / GND / VDD) exposed on a header or bare pads (`J1`), not populated in production, only used during development/flashing.
- **Enclosure**: not yet defined — handlebar mount, outdoor-rated (vibration, weather), sealing still to be designed (see specs doc, "Open points").

This is a summary view; see `gopro_garmin_remote_specs.md` §2–3 for the full functional/technical spec and BOM.

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2. Battery vs. programmer: can the CR2032 stay in place while flashing with an ST-Link V2?

Short answer: yes, this works, and it's the normal way to do it — the CR2032 stays as the only power source, and the ST-Link V2 only taps the SWD signal lines, it does not need to (and normally should not) power the board itself.

- **What the SWD connector's "VDD" pin actually does.** On both the nRF52's SWD pins and the ST-

Link V2 side, the pin often labeled `VDD`/`VAPP`/`VDD_TARGET` on this kind of connector is a **voltage *sense* input**, not a power output: the probe reads it to know the target's logic level (for signal-level compatibility / level shifting), it doesn't source current to power the board unless you deliberately wire it that way. ST's own community forum threads on this are explicit: *"The STLINK V2 adapter does not supply your target unless you do it yourself using a separate 3.3V source"*, and *"On a real ST-LINK this is meant to be an input for detecting/measuring board voltage."* Boards that *are* powered by the probe (e.g. some Nucleo setups) do so through a separate, deliberate jumper/wiring choice, not as default SWD behavior. ST community: SWD operation without connecting VDD_TARGET · ST community: power output *to* target?

- **Practical consequence for this board**: wire `J1` as SWDIO / SWCLK / GND, plus the sense line back to the board's own VDD (i.e. the CR2032 rail) so the probe knows it's talking to a ~3V part — do **not** tie that pin to the probe's own 3.3V output. The CR2032 keeps powering the nRF52 exactly as in normal operation; the ST-Link V2 (via OpenOCD, as already noted in the specs doc) just observes/drives the two SWD signal lines.
- **Is a coin cell "enough" to power a board *while* it's being flashed/debugged?** Yes for this use case. The debug link itself adds negligible current (a couple of logic-level GPIOs), so it doesn't meaningfully add to what the CR2032 already has to supply when the radio (BLE/ANT+) is active — which the buffer capacitor is already sized for (see specs doc, "Power supply"). The only caveat: a partly depleted cell with high internal resistance could sag under a flash operation's brief current pulses the same way it could during normal radio TX — using a reasonably fresh cell for bring-up/debug sessions avoids flakiness that would otherwise be misdiagnosed as a firmware or SWD-link bug.
- **J-Link EDU Mini vs. ST-Link V2**: the same reasoning applies to either probe (per the specs doc, J-Link EDU Mini is the primary option, ST-Link V2 + OpenOCD the noted alternative) — both are sense-only on the target-voltage pin by default.

Conclusion: no hardware change needed to support "flash while running off the coin cell" — this is already what the existing `J1` SWD header (per the specs doc BOM) is for. Worth adding to the PCB/BOM notes explicitly so nobody wires VDD as a power feed by mistake later.

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3. Reverse-polarity protection for the CR2032

Question: does this design need explicit reverse-polarity protection (diode or MOSFET) on the coin cell, or is the mechanical protection from the holder itself (most CR2032 holders only accept the cell one way) sufficient — and what do comparable designs actually do?

Findings (not yet a final decision — see recommendation below):

- **Diode protection is a poor fit for a CR2032 design.** A series diode's forward drop (~0.3–0.7V depending on type) eats directly into the CR2032's already-thin 3.0V→~2.0V discharge curve. Community discussion on exactly this trade-off (Nordic DevZone, for an nRF51822/coin-cell product) frames it the same way this project would need to: a diode effectively raises the *usable* end-of-life voltage from ~2.0V to ~2.7V, which can throw away a meaningful fraction of the cell's usable capacity — a real problem for the "several months on one CR2032" target already stated in the specs doc. Nordic DevZone: MOSFET or Schottky diode for reverse battery protection with a coin cell + nRF51822
- **A P-channel MOSFET is the standard low-loss alternative**, when protection is wanted at all: wired so its body diode/gate-source relationship only turns it on for correct polarity, its on-resistance drop is a small fraction of a diode's forward drop (single-digit mV to low tens of mV at these currents, depending on the part), which is much more compatible with a coin-cell power budget.

components101: PMOS design guide for reverse-polarity protection · Hackaday: reverse voltage protection with a P-FET

- **Correction vs. the earlier draft of this section — "mechanical keying" is weaker than it sounds for a plain CR2032.** A CR2032 is a **symmetric disc** (20mm diameter on both faces, only the polarity marking differs) — it is not "keyed" by its own shape the way, say, a polarized connector is. Generic open-frame/spring-clip holders (the common, cheap kind, including the through-hole part picked in §4 below) mechanically accept the cell **either way round**; only a dedicated anti-reversal mechanism (e.g. a housing shaped so the cell only seats correctly one way, the subject of at least one US patent specifically on this problem) truly blocks reversal, and that is not what most inexpensive holders implement. Forum discussion of this exact question (Arduino Forum, ST community, Contextual Electronics Forum) confirms the same conclusion: physical reversal is generally *possible* in a standard holder, so "mechanical protection" in practice means clear +/- silkscreen/labeling plus the mild inconvenience of doing it wrong, not a hard mechanical block. Arduino Forum: reverse polarity protection for coin cell? · ST community: typical practice for protecting VBAT from a reversed coin cell · Contextual Electronics Forum: CR2032 reverse polarity protection? · US7118817B2 — coin cell protection against reverse insertion in cell holder
- **Other real designs go either way depending on risk tolerance / target market**, not a single universal answer:
  - Some nRF52 + coin-cell reference designs add explicit reverse protection (one example found uses an LDO plus a dedicated reverse-polarity protection diode ahead of it) — but that's in a *regulated* design, where the diode's drop is absorbed by the LDO's headroom, a very different power budget from this project's *unregulated*, direct-to-chip CR2032 supply. Northwest Engineering Solutions: nRF52 module with battery power
  - TI's own CR2032-powered wireless-sensor reference designs (e.g. MSP430FR5969 / HDC2010-based designs cited above) are built the same way this project is (direct-to-chip, no regulator) and do not appear to add a discrete reverse-protection component in their BOM — consistent with relying on the holder's mechanical keying. TI TIDUDW6 reference design · TI TIDUDW5 reference design

Recommendation for this project: given the direct-to-chip (unregulated) power architecture and the "several months on one CR2032" target, a series diode is not a good fit (voltage-budget cost too high). Given the correction above — the through-hole holder selected in §4 does **not** mechanically block reversed insertion — "mechanical-only" is not actually a safe option here on its own:

1. ~~**Mechanical-only**~~ — ruled out: the holder chosen in §4 (like most generic open-frame CR2032 holders) physically accepts the cell both ways round, so this would mean shipping with no real protection beyond a silkscreen marking.
2. **Add a P-channel MOSFET** in the battery-negative or -positive leg for near-zero-drop protection — a few extra cents/mm² of PCB space. **This is the recommended option**, specifically because the chosen holder (§4) is not self-keying and the battery is user-replaceable (a rider will swap the CR2032 in the field, without the assembly-line care a sealed/factory-only device might get).

Decision: add the P-MOSFET reverse-polarity protection stage to the BOM/schematic. This supersedes the earlier "open, pending holder choice" framing now that §4 has picked a specific (non-keying) holder part.

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4. CR2032 through-hole holder selection

Requirement: a through-hole (solder-in-PCB) CR2032 holder, easily sourced from standard distributors — and, per user correction, the cell must sit **flat against the PCB** (lying parallel to the board, behind the Ebyte

module), not standing on edge or protruding vertically, since this has to fit in a slim handlebar-mounted enclosure.

Correction — the Keystone 1074 (previously recommended) does not fit this constraint. It is a "top-loading" holder: the cell sits with its flat faces roughly perpendicular to the PCB and is dropped in from above, with the manufacturer's own listing giving a **0.72" (18.3mm) height above the board** — essentially another cell's diameter of clearance standing straight up off the PCB. That's fine for a project with a deep case (or the battery on a separate compartment) but wrong here, where the datasheet directly says the cell must lie flat behind the Ebyte module in a slim enclosure.

Revised recommendation: MPD (Memory Protection Devices) `BC2032-E2` — a horizontal, through-hole, "open" coin-cell holder where the CR2032 lies flat in the plane of the PCB.

- **Through-hole / PC-pin mount, low profile**: **5.08mm height above the board** (per DigiKey's listing) — only slightly more than the CR2032's own 3.2mm thickness, consistent with "flat behind the module," not standing proud of the board. DigiKey: BC2032-E2
- **Widely available / not single-sourced**: stocked at DigiKey, LCSC, JLCPCB's parts library (convenient if boards are ever assembled there), and listed by several other distributors under the same MPD part number. LCSC: BC2032-E2 · JLCPCB parts: BC2032-E2 · MPD/batteryholders.com: BC2032-E2
- **Advertised "reverse polarity protection"**: several distributor listings describe this specific part as including reverse-polarity protection, which — if real and mechanical rather than just a marketing label — would be a nicer property than the generic open-frame holders discussed in §3. **Not independently verified here**: no source found actually diagrams *how* the mechanism works (contact shaping, an insulating step, etc.), so treat this as an unconfirmed bonus, not a substitute for the P-MOSFET protection decided in §3 — confirm from MPD's actual datasheet/mechanical drawing before relying on it, and keep the MOSFET regardless.
- Accepts the wider 20mm coin-cell family (CR2032 and similarly-sized cells), same as most parts in this class.

Alternative (thicker, but with a more clearly documented mechanical outline): NTE Electronics `23-BHCC-1` — also horizontal/through-hole, "Low Profile for High Density Packaging" per NTE's own datasheet, 9.2mm height above board (roughly double the BC2032-E2's), stocked at DigiKey/Hawk Electronics/Micro Center. Worth keeping as a fallback if the BC2032-E2 ever goes out of stock or its reverse-protection claim doesn't check out against the real datasheet. DigiKey: 23-BHCC-1 · NTE datasheet (coin-cell holders)

Still applies from §3: whichever of these is used, treat it as "polarity-marked," not "polarity-proof" (pending the BC2032-E2 datasheet check above) — the P-MOSFET protection stage stays in the BOM.

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5. Buttons — size, sealing, sourcing

Requirement (per project owner): 4 or 5 momentary push buttons (2 for the camera, page-left/page-right for the Garmin, optionally a 5th — see "Handling of the 2 camera buttons" in the specs doc's open points for whether camera ON/OFF stays 2 separate buttons), each:

- large enough to operate **with cycling gloves on** — actuator diameter roughly **7–10mm**, possibly a bit less,
- **rain-resistant** (splashing/dripping water while riding, not submersion),
- **easy to source and cheap**.

Rain-resistant vs. waterproof — refining the IP target. "Rain on a handlebar" is the IEC 60529 **IP65**

condition (dust-tight + protected against low-pressure water jets from any direction), not the stricter **IP67**/**IP68** (temporary/continuous full immersion) that most "waterproof switch" marketing copy leads with. Any switch rated IP65 or better covers the actual requirement here — IP67-rated parts (common and not meaningfully pricier in this switch class) are fine too, just not something to pay extra to chase beyond IP65.

Revised requirement (user correction #1): PCB-mount, not panel-mount. The panel-mount switches originally proposed here (bolted through the enclosure wall, wired to the board) are the wrong shape for a genuinely compact design — they need standoff depth behind the panel for the switch body/bushing/nut, and separate wiring. **Soldered directly onto the PCB** is both more compact and simpler to assemble; that constrains the choice to a *sealed tactile switch*, not a generic panel-mount button.

Revised requirement (user correction #2): through-hole, not SMD. The first PCB-mount candidate proposed here (APEM/MEC's `Ultramec 6C`) was surface-mount — ruled out, since this project's hand-solder-with-an-iron approach (already the rationale for the Ebyte module's extended-pad footprint in §1) favors through-hole parts. That rules out most "low-profile sealed keypad switch + floating cap" families, which tend to be SMD-first (including Ultramec 6C and C&K's KSE mentioned in an earlier revision of this section) — but not all of them.

Recommendation: APEM/MEC `Multimec 5G` series, through-hole, with a round "floating cap" accessory.

- **Switch itself**: **through-hole**, IP67 sealed to IEC 60529 (PPS housing + actuator, **silicone rubber sealing**, stainless steel contact springs) — a genuinely sealed switch, not "sealed by an external boot." Base footprint **10mm x 10mm, 6.4mm body height**, rated **10 million actuations** on the general 5G platform (a specific through-hole/cap combination found at RS is speced to 500,000 — check the exact ordering code chosen against its own datasheet page, cycle rating varies by variant). Mouser: Multimec 5 series datasheet — 10M actuations, IP67 sealing · TTI: APEM Multimec 5G series datasheet
- **What reaches the 7–10mm actuator target without an SMD floating cap**: the 5G platform takes the same "separately-molded oversized cap on a compact sealed switch" approach as the SMD Ultramec 6C considered earlier, just on a through-hole base — e.g. cap code **`1ES`, a round cap Ø9.6mm**, sitting right inside the requested 7–10mm window, or the plain round polyamide cap seen in a stocked RS variant at **Ø12mm** (slightly larger) if a bigger target is preferred. Total stack height (switch body + this class of cap) is on the order of **~12.5mm** per the datasheet — noticeably more than the bare 2.5–6.4mm switch body, since a sealed, glove-sized cap necessarily adds some stack-up; treat this as the real constraint on enclosure thickness behind each button, not the switch body alone. Enika: Multimec 5G through-hole dimensions incl. 1ES cap
- **Widely available, in through-hole, at mainstream distributors**: confirmed stocked at **RS Components** (e.g. `5ETH935+1SS09-12.0` with the 12mm polyamide round cap, IP67, silicone-sealed, 2mm recommended panel thickness), with the wider Multimec 5G platform also documented via Mouser/Farnell/TTI datasheets — this is a standard, multi-sourced industrial switch family, not a boutique or single-distributor part. RS: MEC IP67 round-button tactile switch, 12mm, through hole
- **Hand-solder compatibility**: through-hole pins, no reflow needed — matches the project's existing iron-solder approach better than the SMD Ultramec 6C did.
- **Bonus for this project's 4–5 button layout**: like the Ultramec 6C, caps come in multiple colors (the RS-listed variants above are black/grey/red/white combinations) — useful for marking the two camera buttons distinctly from the Garmin page-left/page-right/lap buttons without separate labeling.

Why not a smaller, lower-stack sealed THT switch instead: sealed THT tact switches with actuators already in the 7–10mm range and a sub-10mm total stack do not appear to exist as a single stocked part — this class of switch (CIT's `STV` series, C&K's sealed sealed lines, etc.) sticks to the same small-actuator-plus-cap pattern as the SMD world once sealing is required; the Multimec 5G's ~12.5mm stack is close to

the practical minimum for "sealed + through-hole + big enough for a gloved finger" today, not a suboptimal pick within that constraint set.

Open points:

- Final button count (4 vs. 5) — tied to the still-open "Handling of the 2 camera buttons" question in the specs doc.
- Pick the exact cap code (`1ES` Ø9.6mm vs. the Ø12mm polyamide round cap) once the enclosure's front-panel button-hole size is decided, and get a firm per-unit price for the switch+cap combination for the BOM.
- The ~12.5mm total switch+cap stack height now sets a floor on the enclosure's thickness behind each button — fold this into the enclosure design (specs doc §4).
- Real-world IP65/IP67 validation once a prototype board + enclosure exist — with a PCB-mount switch, sealing at the enclosure depends on how the case meets the PCB around each button (a gasket, or the switch's own panel-thickness spec, at the enclosure's button openings).

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6. Alternative: membrane keypad (ZX81-style)

The comparison is apt. A membrane keypad is exactly the technology behind the ZX81's flat keyboard (and calculators, microwave panels, medical-device front panels since): a thin flexible sandwich — usually two printed-circuit polyester layers with a spacer between them, a metal "snap dome" under each key position for tactile click, and a printed graphic overlay on top — instead of discrete mechanical switches. It is a legitimate, very compelling alternative to the Multimec 5G recommendation in §5 on exactly the axis this project cares about: **compactness**.

- **Thickness**: a standard (non-waterproofed) membrane switch stack is on the order of **~0.7mm**; a waterproofed version (thicker overlay + edge-sealing adhesive) is on the order of **~1.4mm** — vs. the **~12.5mm** total stack for a Multimec 5G switch + oversized cap (§5). This is the single biggest advantage: essentially the entire vertical budget a switch+cap combo needs disappears. LID/tactilemembrane: standard vs. waterproof membrane switch stack-up
- **Sealing**: the membrane is bonded to the front panel with a continuous perimeter adhesive (e.g. 3M acrylic adhesive films specifically sold for this), so the "sealing surface" is the whole keypad sheet, not a per-button gasket — **IP65/IP67 is a normal, well-documented outcome** for this construction, not a stretch. LID/tactilemembrane: IP67/NEMA 4X sealed membrane switch design
- **Actuator size/shape is free**: because each "button" is just printed artwork plus a dome underneath, the 7–10mm glove-friendly target from §5 is trivial to hit exactly (or any shape/spacing) — and the graphic layer can print icons/labels directly on each button (camera on/off, left/right arrows, lap) instead of relying on discrete colored caps.

The catch: sourcing is a real trade-off, not a free upgrade.

1. **A genuinely off-the-shelf, catalog-stocked option exists**: **NKK Switches' `FM` series**, e.g. part `FMBN04BE` — a **4-key** non-illuminated membrane keypad, snap-dome tactile feedback, adhesive-backed for panel mounting, 1,000,000+ cycle life, stocked at DigiKey/Mouser/RS/Farnell (not a custom order). This is the closest thing to "buy a membrane keypad the same way you'd buy a switch." DigiKey: FM series membrane keypads · DigiKey: FMBN04BE · NKK FM series datasheet (Mouser-hosted PDF)
    - **Only comes in 4-key or 16-key layouts** — no 5-key option. A 4-key FM keypad is a clean fit *if* the project settles on 4 buttons (i.e. the still-open "camera ON/OFF as one toggle button"

question in the specs doc resolves that way); if 5 buttons are kept, this specific catalog part doesn't fit and the choice falls back to option 2 below.
    - **Listed as "Matrix Output"** despite only 5 pins for 4 keys — **not yet confirmed** whether that's a true scanned row/column matrix (which would conflict with the specs doc's firmware architecture: "5 touch buttons, individual GPIOs, no matrix," wake via GPIO PORT event) or effectively "4 individual switches + 1 shared common," which would be a drop-in fit for the existing GPIO/pull-up design. **Must check the actual pinout in NKK's datasheet before adopting this part** — this is the single biggest open question standing between this option and a firm recommendation.
    - **Termination is a cable with connector**, not bare solder pads — so unlike a discrete through-hole switch, this needs a mating connector footprint on the PCB (a small added BOM line), not a direct hand-solder joint. Still compact and still avoids SMD switch parts, just a different kind of PCB interface than §5's switches.
    - **Operating temperature -15°C to +50°C** (per search results, unconfirmed against the full datasheet) — worth double-checking against realistic cold-weather riding conditions as part of verifying this part.

2. **A fully custom membrane keypad**, made to this project's exact 4-or-5-button layout/graphics by a specialty manufacturer (e.g. RSP, LID/tactilemembrane, Kingway USA, JRPanel, ALLPCB) — the "real" ZX81-style approach: any button count, shape, icon set, and IP rating, and cheap per unit at volume. But it is genuinely a **custom-manufactured part, not a catalog purchase**: typical MOQ for a standard custom design is **~50 units** (some vendors offer 5–10-piece prototype/first-article runs), with **2–8 week** production lead time (a few vendors advertise 24–48h turnarounds specifically for early prototypes), plus one-time artwork/tooling setup. That's a real mismatch for "build one or two units to test," though very attractive once the design is stable and a small production batch is being planned. JRPanel: membrane switch prototype quoting · RSP: custom membrane switch manufacturing · LID/tactilemembrane: custom membrane switches, 2-week prototype

Recommendation: keep §5's `Multimec 5G` (through-hole switch + cap) as the pick for the **first prototype board** — it's a buy-today, solder-today catalog part with no MOQ/lead-time risk, even though it costs ~12mm of stack height. Treat the membrane keypad as the **strong follow-up option once the design stabilizes**:

- If the button count settles at exactly 4, revisit the NKK `FM` series (`FMBN04BE`) as a possible catalog drop-in — pending the matrix-output pinout check above.
- If a small production run (tens to low hundreds of units) is ever planned, a fully custom membrane keypad is likely the better long-term choice on compactness, sealing, and per-unit cost alike — revisit once the enclosure and final button layout are locked.

Open points:

- Confirm `FMBN04BE`'s actual pinout (true matrix vs. common+4) against NKK's full datasheet before treating it as firmware-compatible.
- Resolve the 4-vs-5-button question (specs doc open point) — it gates whether the off-the-shelf NKK 4-key part is even usable.
- If pursuing a custom membrane keypad later, get concrete MOQ/lead time/price quotes from 2–3 of the vendors above once the layout is final.

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7. Temperature sensor — does idle operation make it read true ambient?

Question: since the device is idle most of the time, and the nRF52832/nRF52840 both have a built-in die-temperature sensor, should that reading effectively track outside/ambient temperature?

Partial confirmation — idle operation solves the *electronic* self-heating problem, but that's not the only error source.

- **What idle operation actually fixes**: yes, being idle almost all the time removes the dominant *electronic* self-heating concern. Nordic's own DevZone guidance on this exact question notes the nRF52832 is "an ultra low power device and produces very little heat, so the die temperature is approximately the same as the ambient temperature," citing a measured **~2°C rise on the older nRF51 under close-to-full activity**, with the nRF52 expected to run cooler still for the same workload thanks to its better power efficiency. This project's duty cycle — brief BLE/ANT+ bursts on a button press, then long idle stretches, with temperature sampling only every 5 minutes during an already-bounded broadcast session (see specs doc "Thermometer") — is far below "close to full activity," so self-heating from the radio/CPU itself should be a minor contributor, consistent with what the specs doc already states. Nordic DevZone: nRF52832 temperature sensor accuracy
- **Minor discrepancy to resolve**: that DevZone thread cites **±5°C** stock accuracy, while the existing specs doc's "Internal temperature sensor" section cites **±4°C** "per Nordic's datasheet" — worth reconciling against the actual nRF52832 datasheet's `TEMP` electrical characteristics table (the two figures may reflect different conditions, e.g. calibrated vs. raw, or a different temperature range) before quoting a single number with confidence.
- **What idle operation does *not* fix: the sensor still reads the chip/PCB/enclosure temperature, not free air — and this is where most of the real-world error comes from in comparable devices.** This is a well-documented issue on Garmin Edge and Wahoo ELEMNT units (both of which read temperature the same architectural way: an internal sensor on the board, not an external probe), independent of how "busy" the unit's electronics are:
    - Garmin Edge forum reports describe units in direct sun reading **5–10°F (~3–6°C) higher** than an identical unit in the shade, and up to **10–15°F (~5.5–8°C) higher** in full sun with low wind.
    - Wahoo's own support documentation states the temperature sensor "is placed where it best reads the temperature of the internal electronics, rather than the ambient external air temperature," and that Wahoo considers **±10°F (~5.5°C)** to be within spec — and explicitly calls out direct sunlight, holding the unit in hand, or charging from an external battery as things that push the reading above true ambient.
    - Both sources agree the reading is "mostly accurate... at high airspeed and under overcast conditions" — i.e. exactly the opposite of a stationary, sunny handlebar mount.
    - **Directly answering "so shade would be fine?"**: yes — the same reports describe both Garmin and Wahoo units as accurate to **within ~1°F (~0.5°C) in the shade**, vs. the 5–15°F (~3–8°C) over-read cited above once the unit sits in direct sun. Shade removes essentially all of the *dominant* error source for this class of device; what's left is closer to the sensor's own stock tolerance (±4–5°C, still unresolved above) rather than the much larger solar-gain effect. Wahoo: Data Accuracy — temperature sensor placement and tolerance · Garmin Forums: Edge 1030 Plus temperature reading discussion · Bike Forums: the Garmin Edge thermometer

Net answer: idle operation is genuinely the right reasoning for *why self-heating from the chip itself won't be a big factor* here — that part is confirmed, and it's a meaningfully stronger position than "any die-temperature sensor is unreliable," since this project's duty cycle is much lighter than the "close to full activity" case Nordic measured ~2°C for. But it doesn't make the reading equivalent to true outside air temperature: the sensor still measures a small sun-exposed plastic enclosure sitting still on a handlebar, and that enclosure's own thermal behavior (solar gain, low airflow when stationary, proximity to a warm hand/body) is — per real-world data from architecturally identical commercial devices — good for several °C

of error on its own, on top of the sensor's own ±4–5°C stock tolerance. This matches, and now quantifies with real precedent, the specs doc's existing framing: **fine for an indicative "feels like" reading on the Edge, not for precision measurement** — no change to that conclusion, just firmer evidence behind it.

Open points:

- Reconcile the ±4°C vs. ±5°C accuracy figures against the nRF52832 datasheet directly.
- Check `ant_garmin.c`'s actual session timing: is the *first* temperature sample of a session taken immediately when the triggering button press starts the session, or only after the first full `TEMP_BROADCAST_INTERVAL_MIN` has elapsed? The former would sit closer in time to that button's own BLE/ANT+ activity (a much smaller effect than the enclosure/solar one above, but easy to avoid if it's cheap to delay the first sample slightly).
- Once real hardware exists, field-test the "feels like" framing directly: compare the device's reading against a known-good ambient reference in both shaded and direct-sun handlebar mounting, similar to the Garmin/Wahoo reports cited above.

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8. User-feedback LEDs on an unregulated, decaying CR2032

Question: one or two indicator LEDs (red/green) would help the user (action confirmed, failure, low battery). How do we drive them when the supply is the bare coin cell, whose voltage drifts from ~3.0V (fresh) down to ~2.0V (end of life) and sags further under load?

All figures below were checked against the datasheets linked in §8.8 (values read from datasheet *graphs* are marked "≈ graph").

8.1 The cell voltage is flatter than it sounds

The Energizer CR2032 datasheet shows the Li/MnO_2 discharge curve is a **plateau around 2.9V (background load)** / **~2.8V (6.8mA pulse load)** for roughly the first 75–80% of capacity, then drops steeply towards the 2.0V cut-off (235mAh rated to 2.0V). The same datasheet's IR chart gives an internal resistance of **≈ 10Ω fresh, ≈ 20–25Ω at ~175mAh, then rising past 100Ω** in the last ~30mAh (≈ graph).

Consequence: a plain series resistor sized for ~2.8V gives essentially constant brightness for most of the cell's life. The "varying voltage" problem is real only near end of life — where a dimming LED coincides with the low-battery warning anyway.

8.2 Colour choice: forward voltage

Part (Kingbright, 0805)	**Colour / material**	**Vf typ @20mA**	**Vf at 5–10mA (≈ graph)**	**Iv typ @20mA**	**Usable on CR2032?**
APT2012SURCK	Hyper red 630nm, AlGaInP	1.95V	≈ 1.8–1.85V	230 mcd	**Yes**
APT2012CGCK	Green 570nm, AlGaInP	2.1V	≈ 1.95–2.0V	50 mcd	**Yes**, fades below ~2.2V
APT2012SGC	Green 568nm, GaP	2.2V	≈ 2.0V	12 mcd	Yes, but very dim
APT2012ZGCK	Green 525nm, InGaN	**3.3V**	—	550 mcd	**No** — Vf above the cell voltage

- "Green" must be a **yellow-green AlGaInP (≈570nm)** die, not InGaN: the InGaN part needs 3.3V typ,

more than a CR2032 ever delivers. Many red/green bicolour LEDs use InGaN for the green — check Vf.
- Even the best usable green (CGCK, 50 mcd) is **~4.6× less luminous than the red** (230 mcd) at the same current (mcd already includes the eye's sensitivity). The green needs more current than the red to look comparable.
- Luminous intensity is **linear with current** on both parts (≈ graph), and temperature-dependent: ≈ ×1.5 at –20°C, ≈ ×0.7 at +60°C relative to 25°C (≈ graph). Vf has a –1.9mV/°C tempco, so **+≈ 85mV at –20°C** — slightly less headroom in winter.

8.3 How much current? (≈ 1–3mA is too little)

An earlier draft of this section targeted 1–3mA. That is **too low for outdoor use**: from the datasheets' linear Iv(I), 2mA gives ≈ 23 mcd (red) and ≈ 5 mcd (green). That is fine indoors but marginal for a handlebar in daylight, especially the green. These parts are specified at 20mA.

What limits the current is not energy (a 10mA × 100ms flash costs ≈ 0.28µAh — 1000 flashes ≈ 0.12% of 235mAh) but:

1. **GPIO drive** (nRF52832 PS v1.9 §20.4 *Electrical specification*, p. 158, high drive):
    - VDD ≥ 2.7V: **IOL,HDH 6** / **10** / **15mA** (min/typ/max) at VSS+0.4V; IOH,HDH 6 / 9 / 14mA at VDD–0.4V.
    - VDD ≥ 1.7V: only **3mA guaranteed** (IOL,HDL / IOH,HDL) at 0.4V drop.
    - Standard drive: 1 / 2 / 4mA — unusable here.
    - Nordic (DevZone, Nordic staff) recommends **≤ 15mA total sink/source for the whole chip**, which may all go to one pin, and recommends **sinking** rather than sourcing (more GND pads than VDD pads). → An effective pin resistance of ≈ 40Ω typ / ≈ 67Ω worst case (0.4V / 10mA, 0.4V / 6mA) at VDD ≥ 2.7V is a reasonable back-of-envelope model.
2. **Cell sag**: 10mA × 10–25Ω = 0.1–0.25V for most of the life. On a worn cell (IR > 100Ω) it reaches > 1V.

Recommended target: one LED at a time, peak ≈ 10mA for the green and ≈ 5mA for the red (≈ 25 mcd green, ≈ 58 mcd red — closer in perceived brightness than equal currents). This stays inside the guaranteed high-drive spec and the 15mA chip budget. Going beyond (15–20mA) means an external logic-level N-MOSFET with Rds(on) specified at Vgs ≤ 1.8V, to be justified only if the outdoor test (§8.7) shows 10mA isn't enough.

8.4 Sizing (direct GPIO drive, LED sinking into the pin)

`R = (V_design - Vf) / I - R_pin`, V_design = 2.8V (plateau under load), R_pin ≈ 40Ω typ:

LED	Target I	Vf	R_total	R (E12)	I worst-case pin (67Ω)	I at VDD = 2.4V	I at –20°C
Green CGCK	10mA	2.0V	80Ω	**39Ω**	≈ 7.5mA	≈ 5mA	≈ 9mA
Red SURCK	5mA	1.8V	200Ω	**150Ω** (+ R_pin ≈ 40–50Ω at 5mA)	≈ 4.6mA	≈ 3.2mA	≈ 4.6mA

(Derived values — first-order model, to be checked on the bench.) A fresh, unloaded cell (~3.0–3.2V) pushes these up by ~20–25%, which is harmless (parts rated 30mA DC).

Below ~2.4V the LEDs fade to about half brightness, and the green goes out around 2.1V, which is when the cell is nearly empty anyway (§8.1).

PWM is optional, not required: since the plateau already gives near-constant brightness, PWM is useful mainly to (a) **retune brightness in firmware** after the outdoor test, by sizing R for the maximum wanted current and dimming down, and (b) balance red vs. green by eye. Compensating VDD from the SAADC reading is possible but buys little before end of life.

8.5 Coin-cell interactions

- **The bulk capacitor does not carry a flash**: $\Delta V = I \cdot t / C = 10\text{mA} \times 100\text{ms} / 100\mu\text{F} = 10\text{V}$, so the cell supplies the LED itself. The cap only smooths µs–ms radio bursts (or a PWM ripple at ≥ 1kHz, where the cell sees roughly the average current).
- **Don't overlap the flash with radio TX/RX**: nRF52832 TX at 0dBm draws **11.6mA (LDO)** / **5.3mA (DC/DC at 3V)**, RX 11.7 / 5.4mA (PS v1.9 RADIO electrical specification, p. 234). LED + radio ≈ 15–22mA would sag a worn cell (IR > 100Ω) below 1.7V, the minimum supply voltage (VDD 1.7–3.6V). The brown-out reset is at **1.5–1.7V** (VBOR,ON, PS v1.9 p. 103). Flash *after* the BLE/ANT+ exchange, which is also the right UX (report the result). Whether the chosen module (§9) has the DC/DC (inductor fitted) is to be checked.
- **Measure VDD with the LED off**, or the sag skews the battery-level reading.
- Keep flashes short (50–150ms) and use a blink pattern for "session active" rather than a steady LED.

8.6 Hardware form factor and pins

- **Two discrete 0805 LEDs** (APT2012SURCK + APT2012CGCK), each with its own resistor, **common anode to VDD, cathode → R → GPIO (sink)**: this matches Nordic's sink recommendation, and the LED stays off if the pin floats or is in reset. 0805 is hand-solderable (§1).
- **2-lead anti-parallel bicolour is ruled out at these currents**: the current goes through two GPIO pins in series (2 × ≈ 0.4V at 10mA), so the green would need ≈ 2.0 + 0.8 = 2.8V, the entire plateau voltage. E.g. Kingbright L-59EGW (5mm, GaAsP red 617nm + GaP green 568nm, Vf 2.0 / 2.2V typ, 60 mcd @20mA, per distributor listing) only works at a few mA. A **3-lead common-anode** bicolour would be fine (one GPIO per die, same as discrete) if a single light pipe is wanted, but its green must be AlGaInP/GaP (verify Vf).
- **Pins: avoid P0.22–P0.31.** PS v1.9 §4.3.1 *GPIO located near the radio*, Table 5 (QFN48), p. 18 restricts them to "low drive, low frequency I/O only", which is incompatible with high-drive LED pins. Note that P0.25–P0.31 are also the analog inputs. Also avoid P0.09/P0.10, which are NFC by default (the UICR must be changed to use them as GPIO). Pick from P0.02–P0.08 / P0.11–P0.20, cross-checked against the chosen module's pad map (§9).
- **Zephyr**: plain GPIO → `NRF_GPIO_DRIVE_H0H1` (or `H0S1` for sink-only) in the devicetree `gpios` flags; via PWM/pinctrl → `nordic,drive-mode = <NRF_DRIVE_H0H1>`.
- **Daylight**: a recessed window or light pipe through the sealed enclosure; choose together with the enclosure design.

8.7 Net answer

Red (AlGaInP 630nm) + **yellow-green AlGaInP 570nm** (not InGaN), each with a **series resistor sized for the ~2.8V plateau**, sunk by a **high-drive GPIO**, at **≈ 10mA green** / **≈ 5mA red** peak, one at a time, in short flashes after the radio exchange. No regulator, no boost, no extra IC. The CR2032's flat discharge curve keeps brightness nearly constant for most of its life; PWM is kept only as a firmware brightness knob.

Open points:

- Bench-check the actual currents (R_pin model) and an **outdoor daylight test** through the enclosure window; move to a MOSFET switch only if 10mA isn't visible enough.
- Confirm whether the chosen module (§9) has the DC/DC inductor fitted (radio current 5 vs. 11mA on the nRF52832). If an nRF52840 module is chosen instead, redo the GPIO figures with the nRF52840 PS.
- Choose GPIOs on the chosen module's footprint (§9), outside the radio-restricted pins; add LEDs and resistors to the KiCad schematic (`hardware/board`).
- Define the blink vocabulary in the specs doc (e.g. green = OK, red = failure/no link, red double = low battery). This closes the specs doc's "User feedback (LED/buzzer)" open point on the LED side.

8.8 References

- Nordic nRF52832 Product Specification **v1.9**, local copy `doc/data/nRF52832_PS_v1.9.pdf` — §4.3.1 *GPIO located near the radio* (Table 5, p. 18), §20.4 GPIO *Electrical specification* (p. 158), POWER VBOR (p. 103), RADIO current consumption (p. 234) · online version. The same GPIO / radio / VBOR values appear unchanged in PS v1.0.
- Nordic DevZone, staff answers on GPIO current: max GPIO current for nRF52 (15mA total, sink preferred) · GPIO sink current on nRF52832 (15mA max = at 0.4V drop, 10mA typ)
- Energizer CR2032 datasheet — discharge curves, pulse test (400Ω, ≈ 6.8mA @2.7V), IR vs. capacity
- Kingbright datasheets: APT2012SURCK (hyper red AlGaInP) · APT2012CGCK (green AlGaInP 570nm) · APT2012SGC (green GaP) · APT2012ZGCK (green InGaN, counter-example) · L-59EGW bicolour (element14 listing)
- Zephyr nRF drive-mode macros: `nordic-nrf-gpio.h` · `nrf-pinctrl.h`

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9. Module choice — the E73-2G4M08S1E is an nRF52833

Finding: the module this project was built around, the **Ebyte E73-2G4M08S1E**, is **not an nRF52832**. Its own user manual (v1.9, 2026-03, local copy `doc/data/E73-2G4M08S1E_UserManual_EN_v1.9_fr.pdf`) states "nRF52833 … nRF52833-QIAA / aQFN73" (pp. 2 and 4). Its E73 series table (p. 11) lists the nRF52832 variant as a different part, the **E73-2G4M04S1B**. The 13 × 18mm, 43-pad description in §1 and in the specs doc matches the S1E. The "nRF52832" label attached to it is what was wrong.

Why it matters: the ANT add-on for nRF Connect SDK lists only the **nRF52832 and nRF52840** in the nRF52 series (sdk-ant v2.0.0 / v2.1.x, compatibility table). The **nRF52833 is not listed**. It might work, but nothing confirms it, and ANT+ is a core requirement. **Decision: the nRF52833 / E73-2G4M08S1E is dropped (too risky).**

9.1 Hard requirement found on the way: a 32.768kHz crystal

The nRF52832 PS v1.9 (§19.4.3–19.4.4, p. 112) gives:

Parameter	Value
LFRC (internal RC) tolerance after calibration, `fTOL_CAL_LFRC`	±250 ppm
LFXO tolerance required by the **BLE** stack, `fTOL_LFXO_BLE`	±250 ppm
LFXO tolerance required by the **ANT** stack, `fTOL_LFXO_ANT`	**±50 ppm**

→ BLE alone can run on the calibrated RC oscillator, **ANT cannot**: it needs a 32.768kHz crystal (LFXO). A DevZone answer about ANT on the LFRC sums it up as "may work in laboratory environments, but will

probably fail in the field" (DevZone: using ANT with LFRC). **Any candidate module must therefore have the 32.768kHz crystal fitted, or at least expose XL1/XL2 so one can be added on our board.**

9.2 Candidates

An external 32.768kHz crystal is acceptable. It costs one crystal (±20–50 ppm) and two capacitors on our board. The module only has to expose XL1/XL2 (P0.00/P0.01). Most third-party modules work this way.

Module	**SoC**	**ANT (sdk-ant)**	**Size (mm)**	**32.768kHz crystal**	**Pads / hand-solder**	**DC/DC inductor**	**Factory firmware**	**Verdict**
Ebyte **E73-2G4M08S1E**	nRF52833	**not listed**	13 × 18	not on module, XL1/XL2 exposed	castellated, 1.27mm	currents quoted "DC/DC, 3V"	none	**dropped**
Ebyte **E104-BT5032A**	nRF52832	listed	**11.6 × 16**	**no, and XL1/XL2 not exposed**	castellated, 1.27mm, 24 pads	probably not (TX 13 typ / 19mA max ≈ LDO-mode figures)	**AT firmware, read-protected**	**ruled out**
Ebyte **E73-2G4M04S1B**	nRF52832	listed	17.5 × 28.7	**built in**	castellated	to check	none	fallback (big)
Ebyte **E73-2G4M08S1C**	nRF52840	listed	13 × 18	**built in**	to check	to check	none	candidate
Fanstel **BT832**	nRF52832	listed	14 × 16 × 1.9	external, **XL1/XL2 on castellated pins 3/4**	**16 castellated pins (iron OK)** + 24 LGA pads underneath (optional)	**inductors inside the module**	none stated	**preferred**
Raytac **MDBT42Q** (-P = PCB antenna)	nRF52832	listed	**10 × 16 × 2.2**	external, **XL1/XL2 on pads 13/14**	**edge pads extend past the outline → iron OK**. Only P0.23/P0.24 sit underneath (can stay unused)	DC/DC is the default mode. The ref. circuit shows L2/L3 → location to check	none stated (an "-AT" variant exists separately)	alternative (hand assembly)
Raytac **MDBT50Q** (-P = PCB antenna)	nRF52840	listed	10.5 × 15.5 × 2.05	external, XL1/XL2 on pads 17/18	**two inner rows of pads → hot air / reflow, not iron**	**Reg1 inductor inside the module**	Raytac test code (erasable)	candidate if hot-air soldering is OK

Sources:

- E73-2G4M08S1E manual (local copy above).
- E104-BT5032A manual v1.5 (PDF, §3–4).
- Ebyte product pages: E104-BT5032A, E73-2G4M04S1B ("Built in 32.768K real-time clock crystal oscillator"), E73-2G4M08S1C ("nRF52840 … 13.0*18.0mm … With 32.768K real-time clock"). The S1B size comes from the series table in the S1E manual (p. 11).
- Raytac datasheets: MDBT42Q (§2.1 pinout, §2.2 recommended pads, §8 reference circuit: "Module's default is using DC-DC mode, and must connect it to external 32.768khz") · MDBT50Q (§2.1–2.2, §8: "32MHz crystal and RF (VDD) DC/DC inductor (Reg1) are already inside the module").
- Both Raytac datasheets ask for a **3.8mm no-ground zone** under the antenna end.
- Fanstel BT832 datasheet v2.04 (PDF): "16 castellated pins … SMT equipment is not required for soldering castellated pins", pin table p. 11, "Two inductors required for DCDC converter are inside BT832 module" and "we suggest adding a 32.768 kHz crystal and 2 capacitors" (p. 18). The antenna area should overhang the host PCB / ground plane by 6mm (p. 12).

E104-BT5032A, details: attractive on size and chip, but:

- **No 32.768kHz crystal.** A DevZone user says so (thread 74531), and it is consistent with the pinout: P0.00/P0.01 (XL1/XL2) are not among the 24 pads (manual §4). One cannot be added, which **rules it out for ANT** (§9.1).
- Ships with Ebyte's serial-to-BLE AT firmware and is read-protected. It needs `nrfjprog --recover` over SWD (pads 23/24). Two DevZone users report trouble (74531, 82036), probably a SoftDevice configured for an absent LFXO.
- Only 8 freely named GPIOs (P0.02–P0.07, P0.25, P0.26). The 9 AT function pads (MOD, WKP, DISC, LINK, DATA, RXD, TXD, RTS, CTS) have no documented nRF pin mapping.

Fanstel BT832 (nRF52832):

- **Watch the variant suffix — only the plain `BT832` is an nRF52832.** The family shares a footprint but not the SoC: `BT832` = **nRF52832** (512KB/64KB, Cortex-M4F with FPU, NFC); `BT832A` and `BT832AF` = **nRF52810** (192KB/24KB, M4 without FPU) — *avoid*: wrong chip, too little flash/RAM for the ANT+BLE SoftDevice, and the nRF52810 is **not in the sdk-ant compatibility list** (only 52832/52840). `BT832F` is also an nRF52832 but a larger long-range (FEM) variant, unnecessary here. The JLCPCB part `C7092400` (§9.5) is the correct plain `BT832`. Mouser: BT832A/AF are nRF52810 · DigiKey: BT832F (nRF52832)
- Same chip as the firmware, the specs doc and §8 assume.
- 14 × 16mm, smaller than the E73.
- **DC/DC inductors confirmed inside the module** (open point §8.5 closed for this module).
- **The 16 castellated pins alone cover our needs**:
  - power and debug: VDD, GND, SWDIO/SWDCLK, P0.21/RESET;
  - crystal: XL1/XL2 (P0.00/P0.01);
  - LEDs: two of P0.02, P0.03, P0.13, P0.18, P0.20, which are high drive and outside the radio-restricted range;
  - buttons: P0.26, P0.27 (radio-near, fine for slow inputs) plus the remaining three of P0.02/P0.03/P0.13/P0.18/P0.20;
  - spare: P0.09/P0.10, once switched from NFC in the UICR. The 24 LGA pads underneath can stay unconnected, so the module can be soldered with an iron.
- **In the JLCPCB parts library** (§9.5).
- Needs the external 32.768kHz crystal, and its PCB antenna should overhang the board edge by

6mm, which constrains the enclosure layout.

Raytac MDBT42Q (nRF52832):

- It is the chip the firmware, the specs doc and §8 were written for, so the §8 GPIO figures stay valid.
- It is the **smallest hand-solderable option** (160mm², vs 234mm² for the E73 13 × 18).
- All 32 GPIOs are brought out. On the left edge: P0.25–P0.31, DEC4, DCC, VDD. On the right edge: P0.11–P0.22, SWDIO/SWDCLK and P0.21/RESET. On the bottom edge: XL1/XL2, P0.02–P0.10.
- Raytac is a Nordic module partner and the module is sold through the usual distributors (Mouser datasheet listing, Tindie).
- It needs the 32.768kHz crystal on our board.

E73-2G4M04S1B (nRF52832): built-in LFXO and no factory firmware, but **~2.1× the area** of the S1E (17.5 × 28.7 vs 13 × 18mm), and a PCB antenna with its keep-out. Only worth it if the MDBT42Q is unavailable.

E73-2G4M08S1C / **MDBT50Q (nRF52840)**: ANT-supported, and the **same chip as the nRF52840 Dongle** used for BLE bring-up. The firmware already plans to support both nRF52832 and nRF52840 (specs doc, "Decision: support both nRF52832 and nRF52840"). The S1C has a built-in LFXO but its pads are not yet checked. The MDBT50Q needs hot-air soldering. Choosing either means redoing §8 against the nRF52840 PS.

9.3 Recommendation

Decision (updated) — three modules kept, in this order:

1. **Fanstel BT832 (nRF52832) + 32.768kHz crystal on our board** — primary. Works for both hand soldering (castellated pins only) and JLCPCB assembly (§9.5, Economic PCBA, no X-ray).
2. **Ebyte E73-2G4M08S1E (nRF52833)** — second choice, **kept because we already have this module in hand** (useful for immediate BLE bring-up and, importantly, to *test whether ANT actually runs on the nRF52833*). This partially reverses the "dropped" verdict at the top of §9: the risk is unchanged — **the nRF52833 is still not in the sdk-ant compatibility list (§9)**, so ANT support stays *unverified*; it is a second choice precisely because that risk is only acceptable on hardware already owned, not for a new build. If the ANT test fails, fall back to option 1 or 3.
3. **Ebyte E73-2G4M04S1B (nRF52832)** — third choice: ANT-supported and with a built-in LFXO (no external crystal needed), but ~2.1× the area of the others (17.5 × 28.7mm).

Previously-listed alternatives now set aside (nRF52840 path, or non-JLCPCB): Raytac MDBT42Q, E73-2G4M08S1C, MDBT50Q. The E104-BT5032A remains ruled out.

Before freezing the PCB:

1. **Footprint** from the BT832 datasheet (castellated pins extended outward for iron soldering, like the current E73 template; LGA pads either omitted or left unconnected). The antenna overhangs the board edge by 6mm.
2. **32.768kHz crystal**: ±20 ppm class, CL compatible with the nRF52832 LFXO (`CL_LFXO` 12.5pF max, ESR ≤ 100kΩ — PS v1.9 §19.4.4), plus its two load capacitors, placed close to castellated pins 3/4. Firmware: `CONFIG_CLOCK_CONTROL_NRF_K32SRC_XTAL=y` and the matching `..._K32SRC_*PPM` option.
3. **DC/DC**: inductors are inside the BT832; enable the DC/DC in firmware (radio current 5.3 vs 11.6mA TX 0dBm, nRF52832 PS).
4. **Pin plan** (castellated pins only): LEDs on two of P0.02 / P0.03 / P0.13 / P0.18 / P0.20; buttons on P0.26, P0.27 and the remaining three. Consistent with §8.6.
5. Bring-up: flash an sdk-ant BLE + ANT sample (e.g. the HRM relay sample) on a BT832 board (e.g. Fanstel's EV-BT832) with the LFXO enabled.

Knock-on edits once the module is confirmed (not done yet): `gopro_garmin_remote_specs.md` (MCU section, BOM, ANT section), `software/gopro_remote_fw/README.md`, the comment in `software/gopro_remote_fw/src/main.c`, `hardware/board/README.md` and the KiCad symbol/footprint. All of them currently say "E73-2G4M08S1E (nRF52832)".

9.4 Other manufacturers (not yet checked)

Nordic keeps an official list of nRF52832 modules (nRF52832 modules), with about 30 vendors. The table below is a shortlist filtered for size ≤ ~16 × 16mm, enough GPIOs (≥ 7 needed: 5 buttons + 2 LEDs) and an "open CPU" listing. Sizes and GPIO counts come from that list. Only the crystal column was checked, and only where a source is given.

Vendor / module	Size (mm)	GPIO	Notes
Raytac MDBT42 (no "Q")	8.8 × 13.8	32	Smaller sibling of the MDBT42Q, pads not checked
Fanstel BM832	10.2 × 15	32	Smaller Fanstel sibling of the BT832 (§9.2), pads not checked
Ezurio (ex-Laird) BL652-SA	10 × 14	32	External 32.768kHz crystal optional (Ezurio FAQ). Industrial-grade, pricier
u-blox BMD-300 / BMD-350, ANNA-B112	14 × 9.8 / 8.65 × 6.4 / 6.5 × 6.5	32 / 32 / 25	**No LF crystal on module** (u-blox AppNote UBX-20009617). Small parts likely LGA (not iron-friendly)
Dynastream D52MD2M8IA	14 × 9.8	24	Dynastream is Garmin's ANT subsidiary, so an ANT-oriented module. Availability not checked
MinewSemi MS50SFA	15.8 × 12	13	Not checked
Holyiot 17095	9.4 × 9.25	11	Cheap (AliExpress), not checked

Ruled out from the same list: AT-command-only parts (e.g. Würth Proteus-I/II, u-blox NINA-B11x "u-connect" variants), and LGA/SiP parts too small to hand-solder (Insight SiP ISP1507, KAGA FEI ES2832, WISOL BTS100A…).

9.5 JLCPCB assembly (PCB + SMT)

JLCPCB can assemble the board if the module is in its parts library (LCSC stock). With that, the hand-solder constraint (§1) only matters for prototypes, and the passives (crystal, capacitors, LED resistors) can be placed by JLCPCB too. Library status checked on the JLCPCB part pages. Stock and price change daily, so check them when ordering.

Module	SoC	JLCPCB part	Library	PCBA type	Notes
Fanstel BT832	nRF52832	C7092400	Extended	**Economic + Standard**	EasyEDA symbol/footprint available
Ebyte E73-2G4M08S1C	nRF52840	C356849	Extended	**Standard only, X-ray required**	Built-in LFXO
Ebyte E73-2G4M04S1B	nRF52832	C411306	—	Economic + Standard	28.7 × 17.5mm, built-in LFXO

Raytac MDBT50Q-512K	**nRF52833**	C6904155	Extended	Economic + Standard, X-ray	**Not usable**: the -512K variant is the nRF52833 (the nRF52840 one is MDBT50Q-1MV2)
Raytac MDBT42Q	nRF52832	not found	—	—	Would need JLCPCB global sourcing or consigned parts (not checked)

"Extended" parts carry a per-part loading fee at JLCPCB (Basic parts don't). One module plus a few passives keeps that small.

9.6 Cost estimate (module placed at JLCPCB)

Rough all-in cost *of the module slot only* (module price + its share of the PCBA overhead), to compare the three realistic candidates. Excludes the bare-PCB fab, the other components, the 32.768kHz crystal + 2 caps needed by the nRF52832 modules (~$0.2–0.5 BOM), and shipping. Figures are indicative (LCSC/JLCPCB, ~Sep–Oct 2025; the BT832 was out of stock at the time of the check — reconfirm in the quote tool at order time).

Module prices (LCSC, per piece):

Module	**SoC**	**LCSC/JLCPCB**	**@1**	**@10/100**
E73-2G4M04S1B	nRF52832	C411306	~$5.03	~$4.59 (100)
Fanstel BT832	nRF52832	C7092400	~$7.37	~$7.19 (10), ~$5.9 (1000)
E73-2G4M08S1C	nRF52840	C356849	~$7.59	~$5.52 (100)

JLCPCB overhead that differs between them:

- **Filière**: the two nRF52832 modules (S1B, BT832) are **Economic PCBA-eligible → no mandatory X-ray**. The nRF52840 **S1C is Standard PCBA only and requires X-ray** (§9.5).
- **Setup fee** (once per order): Economic $8 · Standard single-side $25.
- **X-ray** (per board, S1C only): $1.57 (1–10) · $0.79 (11–50) · $0.47 (51–200).
- **Extended-part loading**: ~$3 once per order (all three are Extended).

All-in per board (setup/X-ray amortised over the lot):

Module	**@5 boards**	**@100 boards**
E73-2G4M04S1B (52832, Economic)	≈ $7.2	≈ $4.7
Fanstel BT832 (52832, Economic)	≈ $9.6	≈ $6.4
E73-2G4M08S1C (52840, Standard + X-ray)	≈ $14.8	≈ $6.3

Reading: in prototype quantities the nRF52840 S1C costs ~2× the S1B to place, driven by the forced Standard PCBA + X-ray, *not* by the chip price; the gap nearly closes at 100+. The BT832 is the dearest of the two nRF52832 options as a bare module (~$7.4 vs ~$5.0), but still well under the S1C in prototype runs because it avoids X-ray — the premium buys a module ~2× more compact (14×16 vs 17.5×28.7mm) than the S1B with the DC/DC inductors already inside.

Sources: JLCPCB C7092400 (BT832) · LCSC C356849 (S1C) · LCSC C411306 (S1B) · JLCPCB PCB assembly price (setup + X-ray tiers).

9.7 I/O (GPIO) selection and pin mapping

We need **7 user I/O**: 5 buttons + 2 LEDs (see §5–§6 and §8), plus the fixed lines (crystal, SWD, RESET, power). This section picks GPIOs that avoid the pins Nordic flags near the radio, and gives the mapping on the pads of each of the three selected modules (§9.3).

Selection rules

- **LEDs → high-drive GPIOs, *outside* the radio-restricted set.** The LEDs are driven at ≈5–10mA from a high-drive pin (§8.3–§8.6), which is exactly the "high current near the radio" case Nordic warns degrades RX sensitivity.
- **Buttons → anything left, radio-restricted pins allowed.** A button is a slow, DC-level input (≤10kHz, the "low frequency I/O" limit), so the restricted pins are fine for buttons — that is where we put them.
- **Avoid the NFC pins P0.09 / P0.10** unless we deliberately switch them to GPIO in the UICR (`CONFIG_NFCT_PINS_AS_GPIOS=y` in Zephyr). They default to the NFC antenna function.
- **Reserved, not usable as user I/O**: XL1/XL2 (P0.00/P0.01, the 32.768kHz crystal — on the BT832 and S1E; the S1B has it internal), SWDIO/SWDCLK (debug), and RESET (**P0.21** on nRF52832, **P0.18** on nRF52833).

Radio-restricted pins ("… low frequency I/O only" in the PS pin table)

- **nRF52832** (BT832, E73-2G4M04S1B): **P0.22–P0.31**. Note P0.25–P0.31 double as the analog inputs. nRF52832 pin assignments
- **nRF52833** (E73-2G4M08S1E) — a *different* list, must not be assumed equal to the 52832's: **P0.02, P0.03, P0.09, P0.10, P0.19, P0.23, P0.25, P0.28, P0.29, P0.30, P0.31, P1.01, P1.02, P1.03, P1.04, P1.05, P1.06, P1.07**. nRF52833 pin assignments

Unified pin plan (one firmware for all three boards)

The intersection of the three modules' broken-out GPIOs is small (the BT832's 16 castellated pins are the limiting factor, and the S1E does not break out P0.27, while its P0.18 is RESET). Exactly **seven** lines are common to all three, which is just enough:

Function	Pin	High-drive?	Notes
LED green	**P0.13**	yes, non-restricted on 52832 **and** 52833	✔ ideal for an LED
LED red	**P0.20**	yes, non-restricted on 52832 **and** 52833	✔ ideal for an LED
Button 1	**P0.02**	input (restricted on 52833, fine as button)	AIN0
Button 2	**P0.03**	input (restricted on 52833, fine as button)	AIN1; also the BT832 AT-mode strap — irrelevant with our own firmware
Button 3	**P0.26**	input (restricted on 52832, fine as button)	—
Button 4	**P0.09**	input	NFC1 → set `NFCT_PINS_AS_GPIOS`
Button 5	**P0.10**	input	NFC2 → set `NFCT_PINS_AS_GPIOS`

Cost of this plan: it spends the two NFC pins on buttons (one UICR setting; we use no NFC). If that is unwanted, use the per-module plans below, which have spare non-NFC pins on the S1E and S1B (only the

BT832, limited to 16 castellated pins, has no alternative to P0.09/P0.10 for the 5th button unless an LGA pad is brought out).

Per-module mapping

Fanstel BT832 (nRF52832) — castellated pins only (LGA pads left unconnected, §9.2). All 16:

Cast. pin	Signal	Role in our design
1	P0.26	Button 3 (radio-restricted, ok for input)
2	P0.27	spare button (radio-restricted)
3	P0.00/XL1	32.768kHz crystal
4	P0.01/XL2	32.768kHz crystal
5	P0.02/AIN0	Button 1
6	P0.03/AIN1	Button 2 (AT-mode strap — n/a for our fw)
7	P0.09/NFC1	Button 4 (needs `NFCT_PINS_AS_GPIOS`)
8	P0.10/NFC2	Button 5 (needs `NFCT_PINS_AS_GPIOS`)
9	VDD	power
10	GND	power
11	P0.13	**LED green** (high-drive, non-restricted)
12	P0.18	spare / alt LED (high-drive, non-restricted)
13	P0.20	**LED red** (high-drive, non-restricted)
14	P0.21/ RESET	RESET
15	SWDCLK	debug
16	SWDIO	debug

Here P0.18 (pin 12) and P0.27 (pin 2) are spares; if we want to free the NFC pins, move Button 4 → P0.18 and keep only one NFC pin (or route an LGA pad). With 5 buttons + 2 LEDs we are at the limit of the 16 castellated pins. BT832 datasheet (pin functions)

Ebyte E73-2G4M08S1E (nRF52833) — plenty of GPIOs, so no NFC pin is needed. Recommended assignment (all high-drive, non-restricted for the LEDs):

Module pad	Signal	Role
11	XL1 (P0.00)	32.768kHz crystal
13	XL2 (P0.01)	32.768kHz crystal
33	P0.13	**LED green** (non-restricted)
32	P0.20	**LED red** (non-restricted)
14	P0.06	Button 1 (non-restricted, could also be an LED)

16	P0.08	Button 2
22	P0.07	Button 3
20	P0.12	Button 4
12	P0.26	Button 5 (non-restricted on 52833)
37 / 39	SWDIO / SWDCLK	debug
26	RST (P0.18)	RESET

(Pads for P0.02/P0.03/P0.25/P0.28–P0.31 and the P1.x lines are the radio-restricted ones on the 52833 — keep them for extra buttons or leave unused. This module can run the unified plan above too, but the plan here avoids burning the NFC pins.) Needs an external 32.768kHz crystal (XL1/XL2, pads 11/13), same as the BT832.

Ebyte E73-2G4M04S1B (nRF52832) — built-in LFXO (no external crystal; XL1/XL2 are not broken out). High-drive pins P0.02–P0.08 / P0.11–P0.20 are plentiful:

Module pin	Signal	Role
28	P0.13	**LED green** (non-restricted)
35	P0.20	**LED red** (non-restricted)
17	P0.02	Button 1
18	P0.03	Button 2
19	P0.04	Button 3
20	P0.05	Button 4
21	P0.06	Button 5
37 / 38	SWDCLK / SWDIO	debug
36	P0.21/RST	RESET

(Avoid P0.22–P0.31 — pins 5–11 and 39–41 — for the LEDs; they are the radio-restricted set and P0.25–P0.31 are also analog inputs. They remain fine as extra button inputs. The NFC pins P0.09/P0.10 are pins 24/25 here and are left free in this plan.)

Common to all three: SWDIO/SWDCLK on `J1` (§2), RESET broken out, LEDs on **P0.13** / **P0.20** (the two pins that are high-drive and non-restricted on both the 52832 and the 52833 — the safe choice across the whole selected set). Firmware pin assignments and the KiCad schematic should follow whichever plan (unified or per-module) is chosen; fold this into the knock-on edits listed in §9.3.

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10. Open points (hardware)

- **Module choice** (§9) — E73-2G4M08S1E (nRF52833) dropped; confirm Fanstel BT832 (nRF52832): footprint (castellated pins), 32.768kHz crystal, pin plan, ANT bring-up, JLCPCB stock at order time; then propagate to specs doc, firmware README/comment, `hardware/board` and KiCad. Alternatives Raytac MDBT42Q, E73-2G4M08S1C, E73-2G4M04S1B.
- **P-MOSFET reverse-polarity stage** (§3) — part selection (Vgs(th), Rds(on) at the CR2032's low

current levels, package size) not yet done; add to schematic/BOM.

- **SWD header (`J1`) wiring convention** — make sure schematic/BOM notes explicitly document "sense-only, board stays coin-cell-powered during programming" (§2) so it isn't miswired as a power line later.
- **Button hardware** (§5–§6) — decide between the `Multimec 5G` through-hole switch+cap (buildable now, ~12.5mm stack) and a membrane keypad (far thinner, but either fixed at 4 keys off-the-shelf pending a pinout check, or custom-manufactured with MOQ/lead-time implications) — final button count, and this choice, both still open.
- **Temperature sensor accuracy figure** (§7) — reconcile ±4°C vs. ±5°C against the nRF52832 datasheet, and confirm the first-sample timing in `ant_garmin.c` relative to the triggering button press.
- **User-feedback LEDs** (§8) — APT2012SURCK + APT2012CGCK candidates (AlGaInP, not InGaN), GPIOs outside P0.22–P0.31, resistors; add to schematic; bench current check + outdoor daylight test.
- Everything else affecting the PCB (enclosure, mounting, sealing) is still tracked in `gopro_garmin_remote_specs.md` §4 ("Open points").